PS2: Parameter-Efficient Fine-Tuning of a Small Language Model with Iterative Improvement

BITS Pilani WILP M.Tech. AIML -- Conversational AI (AIMLCZG521) -- Assignment 2

Name: _____ **BITS ID:** _____ **Date:** 2026-08-12

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0. Domain & Model Choices

- **Domain:** NovaBank retail banking customer support (fictional bank). Chosen because it is one of PS2's suggested domains and matches this project's production background in conversational/voice AI -- retail banking support is a realistic, high-volume, template-heavy customer-support setting where an on-premise assistant's cost/privacy trade-off against a hosted LLM is directly meaningful, not just an academic exercise.
- **SLM (fine-tuned):** `Qwen/Qwen2.5-1.5B-Instruct`. Apache-2.0 licensed, within PS2's 0.5-3B range, strong instruction-following for its size, and well supported by the `transformers` / `peft` / `bitsandbytes` stack this project uses for LoRA/QLoRA.
- **Reference LLM (inference-only, never fine-tuned):** `Qwen/Qwen2.5-7B-Instruct`, run in 4-bit (bitsandbytes NF4). Same model family as the SLM -- this isolates the effect of *scale* rather than confounding it with a different pretraining corpus/tokenizer -- Apache-2.0, ~4.7x larger, satisfying PS2's "7B+" requirement, and small enough to run 4-bit on a single modest GPU.

Task 1 -- Instruction Dataset Preparation (1.5 marks)

(Source: `notebooks/01_dataset_preparation.ipynb`, `data/DATASET_CARD.md`, `img/dataset_stats.png`)

Source, license, generation method. Adapted from `bitext/Bitext-retail-banking-llm-chatbot-training-dataset` (Hugging Face Hub), licensed CDLA-Sharing-1.0 (redistribution/modification permitted with attribution and share-alike terms). The *source* dataset's own generation method, as disclosed by Bitext, is **synthetic + template-based**: expert-curated linguistic templates combined with generative expansion (not scraped real conversations). This derived 338-example dataset is not newly synthesized -- it is built from the source via stratified sampling, placeholder resolution, cleaning, and re-splitting, all scripted and reproducible in `build_dataset.py` (`SEED=42`).

Before/after cleaning counts:

Stage	Rows
Source corpus (full)	25,545
Stratified candidate pool (20/intent x 26 intents)	520
After placeholder resolution (744 `{{SLOT}}` tokens resolved, e.g. all phone-number slots -> `1-800-555-0199`, all URL slots -> `www.novabank.com`)	520
After length filter (empty/truncated response guard)	520 (0 dropped -- source rows already complete)
After duplicate removal	520 (0 duplicates found -- source is pre-deduplicated at the row level)
Final balanced dataset (up to 13/intent, before split)	338
Train / Val / Test	**286 / 26 / 26**

Split and justification. 80/10/10, stratified by intent. With ~338 examples over 26 intents (~13/intent), a larger held-out split (e.g. 70/15/15) would leave too few training examples per intent for a 0.5-3B model to learn intent-specific phrasing; 80/10/10 keeps at least one example of every intent in val and test (used respectively for Task 4 diagnosis and Task 5's automatic metric) while maximising LoRA training data.

3 sample records (see `data/processed/sample_records.json` for the full set):

Intent	Instruction	Response (truncated)
`close_account`	"i want assistancd closing an online account"	"Certainly, I can assist you with closing your online account. To initiate the account closure process, please follow these steps: 1. Log in to your account on our website. 2. Navigate to the account settings..."
`make_transfer`	"wanna perform a bank transfer help me"	"I'm here to assist you with performing a bank transfer... 1. Log in to your online banking or mobile banking app. 2. Navigate to the 'Transfers' or 'Payments' section..."
`check_mortgage_payments`	"where could I check the mortgage payment?"	"I'm here to assist you with checking your mortgage payment... Visit our website at www.novabank.com... reach out to our dedicated customer support team at 1-800-555-0199..."

Dataset quality risks (revisited in Task 4):

1. *Template-driven repetitive phrasing* -- template + paraphrase generation means many same-intent responses share near-identical openings ("I'm here to assist you with...") and numbered-list structure; a fine-tuned model may over-generalise this template even where a shorter answer fits better.

2. *Informal/profane user register* -- 54/338 (~16%) instructions contain mild profanity or frustrated phrasing (intentional in Bitext's design, for tone robustness); responses stay professional throughout, so this should teach de-escalation rather than harm the assistant's register.
3. *Coverage gaps from small per-intent counts* -- ~11 train examples/intent teaches surface phrasing but not every edge case; the frozen benchmark's out-of-scope prompt (B6) specifically probes what happens outside this coverage -- and, as Tasks 4-5 show below, this risk materialized directly.

[Insert `img/dataset_stats.png` and `img/split_distribution.png`]

Task 2 -- Baseline Benchmarking: Base SLM vs. Reference LLM (1.5 marks)

(Source: `notebooks/02_baseline_benchmarking.ipynb`, `results/task2_baseline_responses.json`, `results/task2_side_by_side.csv`)

Frozen 8-prompt benchmark (`data/benchmark/frozen_benchmark_prompts.json`, reused verbatim in Tasks 4 and 5):

ID	Type	Prompt
B1	in-domain factual	How do I activate a new debit card I just received from NovaBank?
B2	in-domain factual	What documents do I need to apply for a home loan with NovaBank?
B3	in-domain factual	An ATM swallowed my card. What should I do?
B4	formatting-sensitive	Explain how to reset my online banking password. Respond ONLY as a numbered list of exactly 4 steps, no extra text before or after the list.
B5	reasoning	I have $500 in my checking account. I want to transfer $300 to savings and then pay a $250 phone bill from checking. Do I have enough balance for both, and in what order should I do them?
B6	out-of-scope	Can you give me a recipe for chocolate chip cookies?
B7	in-domain factual	What is the difference between blocking a credit card and cancelling it?
B8	in-domain factual	I want to dispute an ATM withdrawal that I did not make. Walk me through the process.

All (A) base-SLM and (B) reference-LLM verbatim responses are recorded in `results/task2_baseline_responses.json` / `task2_side_by_side.csv` (both A and B run with identical `GENERATION_KWARGS`: greedy decoding, `max_new_tokens=300`, `repetition_penalty=1.1`).

Limitations / gap table (filled from the actual outputs -- see Notebook 02 for full text):

Dimension	Where SLM (A) failed vs. LLM (B)	Evidence
Domain knowledge	A gives generic, non-NovaBank-branded procedures with no concrete channel details; B consistently grounds answers in NovaBank-specific channels (phone number, novabank.com, "our" branding)	B1, B3
Instruction / format adherence	No gap at baseline -- A already produced exactly 4 numbered steps, no extra text	B4
Completeness / reasoning	A failed the arithmetic task outright: states "you have enough balance for both" then describes paying a $250 bill from $200 remaining -- an internal contradiction, never flags the $50 overdraft. B computes $500-$300=$200, then $200-$250=-$50, correctly concludes insufficient balance	B5
Hallucination	A states a specific "60 days" dispute-resolution window with confident, unhedged phrasing but no source backs that number; B avoids committing to an unverified specific timeframe	B8
Persona / brand voice	A rarely speaks as NovaBank itself ("your bank", "the ATM manufacturer") despite the system prompt instructing it to be the NovaBank assistant; B consistently uses "our", "NovaBank"	B1, B3, B7
Out-of-scope handling	No gap at baseline -- both A and B correctly declined the cookie-recipe request	B6

Inference: the baseline gap is smaller than expected on refusal and strict-format compliance (the 1.5B model already handles those) and concentrated instead on arithmetic/logical reasoning (outright wrong and self-contradictory in A) and consistent NovaBank persona/branding -- making these the clearest fine-tuning targets.

Task 3 -- Parameter-Efficient Fine-Tuning, Iteration 1 (2.5 marks, core)

(Source: `notebooks/03_finetuning_v1.ipynb`, `results/task3_efficiency_summary.json`, `img/v1_training_loss.png`)

Why LoRA is parameter-efficient (intuition tied to the measured numbers below). A pretrained weight matrix already encodes broad language competence; adapting it to a narrow domain does not need a full new matrix of free parameters, because the *change* required typically has low intrinsic rank -- most of the useful adaptation lives in a small subspace. LoRA freezes the base weight W and adds a low-rank correction $W' = W + \frac{\alpha}{r}BA$ where only A (rank $r \times k$) and B ($d \times r$) are trained, $r \ll \min(d,k)$. Applied only to the attention projections (`q_proj,k_proj,v_proj,o_proj`) with $r=8$, this is what produces the measured **0.2446%** trainable fraction below -- and because the frozen base is additionally loaded in 4-bit (QLoRA), the optimizer only needs to track Adam moments for that tiny trainable slice, not the full 1.5B-parameter model, which is what makes fine-tuning fit in well under the ~6GB budget of a modest GPU.

Efficiency evidence (mandatory):

- Trainable parameters: **2,179,072** / Total parameters: **890,795,520** (**0.2446%**)
- Peak GPU memory during training: **5,943 MB**
- Why full fine-tuning would not fit the same budget: Full fine-tuning would make all 890,795,520 base parameters trainable (100% vs. 0.2446% here). Adam's two extra fp32 moment buffers per trainable parameter alone would need roughly 8 bytes x total_params just for optimizer state (~12+ GB for a 1.5B model), before gradients or activations -- multiples of the 5,943 MB peak measured for LoRA + 4-bit above, and would not fit the "modest hardware" budget this project targets.

Training configuration:

Hyperparameter	Value	Justification
LoRA rank `r`	8	Standard, well-tested starting point for <3B models on a few-hundred-example dataset; large enough to capture domain phrasing without overfitting this small a training set.
LoRA alpha	16 (=2r)	Widely-used LoRA-paper default; keeps the effective update magnitude (`alpha/r`) stable if `r` changes later.
LoRA dropout	0.05	Light regularisation against overfitting on only 286 training examples.
Target modules	q/k/v/o_proj	Attention projections capture most of the adaptation benefit at lowest parameter cost.
Learning rate	2e-4	Standard LoRA LR, appropriate since only small adapter matrices are updated.
Epochs	3	Enough gradient signal on 286 examples to fit domain phrasing without memorising verbatim (monitored via val loss).
Batch size (effective)	8 (per-device 4 x grad-accum 2)	Fits in memory with the 4-bit base + LoRA.
Quantization	4-bit NF4, double-quant, bf16 compute	QLoRA-style; frees GPU memory for training activations.
Optimizer	`paged_adamw_8bit`	Pages optimizer state to CPU under memory pressure -- a second safety margin on top of 4-bit loading, for a 6GB-class card.

Training/validation loss (`img/v1_training_loss.png`) decreased smoothly across 3 epochs, eval loss 0.894 -> 0.850 -> 0.839, with no sign of divergence or overfitting blow-up at this epoch count.

[Insert `img/v1_training_loss.png`]

Saved model path: `models/novabank-slm-lora-v1/`

Inference demonstration (prompt B1, freshly reloaded v1 adapter):

> Q: How do I activate a new debit card I just received from NovaBank? v1: "Hello! Welcome to our team of dedicated customer service representatives. We're here to assist you in activating your newly received debit card. Here's what you need to do: 1. Visit our website at www.novabank.com. 2. Log in to your account using your credentials. 3. Navigate to the 'Debit Cards' or 'Account Management' section... call us at 1-800-555-0199. Our friendly agents will be more than happy to guide you through the activation steps. Have a great day!"

Already visibly different from base-SLM (A)'s generic, non-branded answer to the same prompt (Task 2) -- v1 has picked up NovaBank-specific branding, phone number, and persona from the training data.

Task 4 -- Diagnose and Improve, Iteration 2 (2.5 marks, core)

(Source: `notebooks/04_diagnose_improve_v2.ipynb`, `results/task4_*`, `img/v2_training_loss.png`*)*

Diagnosis. Notebook 01's two predicted risks were checked against real v1 output rather than assumed: the length/over-long hypothesis was **refuted** (mean length ratio over a 15-example validation sample = 0.99, not systematically >1.5) and the B4 strict-format hypothesis was **refuted** (v1's B4 response is exactly 4 numbered lines, no extra text). Two different, confirmed weaknesses emerged from reading the full 8-prompt v1 output instead:

1. **Out-of-scope refusal regression (B6).** v1 fully answers the off-topic cookie-recipe request with a complete 8-step recipe instead of declining as the system prompt requires -- a regression from Notebook 02's baseline (A), which correctly declined this exact prompt. Fine-tuning on an all-in-domain, zero-refusal dataset (338 examples, no declines) taught v1 to always attempt a detailed answer regardless of topic.
2. **Arithmetic/reasoning failure (B5).** Asked whether $500-$300-$250 is affordable and in what order, v1 walks through *navigation steps* for both transactions but never computes the intermediate balance or states whether the bill is affordable -- the same failure mode baseline (A) showed.

Change log. *Honest note:* the format-focused change was drafted from Notebook 01's predicted risk before v1 existed to diagnose against; the diagnosis above then refuted that v1 has a format problem. Kept and reported plainly rather than silently dropped, since it had already been run.

#	Diagnosed weakness	Change	Rationale
1	*(refuted by diagnosis)* v1 ignores the strict format instruction (B4)	Add 20 format-focused training examples (existing instructions + prose-free numbered-list responses)	Teaches the "numbered list only" pattern the base data under-represents. In hindsight not warranted; kept as a defensive/reinforcing change
2	v1 answers out-of-scope requests instead	Add 15 out-of-scope refusal examples (non-banking	The 338-example set has zero out-of-scope examples -- the model has never seen a

#	Diagnosed weakness	Change	Rationale
	of declining (B6)	instructions + polite decline-and-redirect responses)	"decline" response to imitate

Neither change targeted Weakness 2 (reasoning) -- that was diagnosed after the change log and retraining were already committed by Notebook 04's pre-wired design; carried forward as a v3 candidate rather than retrofitted here, to avoid changing three things in one iteration.

v2 training: same LoRA config as v1 (r=8, alpha=16, dropout=0.05, 3 epochs, batch 4 x accum 2), only the training set changed (286 -> 321 examples: +20 format, +15 refusal). Eval loss 0.888 -> 0.845 -> 0.837, comparable convergence to v1. Saved to `models/novabank-slm-lora-v2/`.

[Insert `img/v2_training_loss.png`]

v1-vs-v2 comparison (before/after per diagnosed weakness):

Weakness	v1 (before)	v2 (after)	Verdict
Format adherence (B4)	Correct 4-item list	Correct 4-item list	**Unchanged** (never actually broken)
Out-of-scope handling (B6)	Full cookie recipe (wrong)	Full cookie recipe (still wrong)	**Unchanged -- targeted fix did not work.** 15/321 (4.7%) refusal examples were not enough signal against 306 examples all modeling "answer fully"
Reasoning (B5, not targeted)	Wrong / self-contradictory	Correct: $500-$300=$200, $200-$250=-$50, correctly flags insufficient balance	**Improved, unexpectedly** -- no change in this iteration targeted reasoning; likely incidental, not claimed as reproducible

Did anything get worse? Yes, honestly reported: **B2** (home loan documents) regressed -- v1 gave a direct 7-item document checklist; v2 instead asks the user to "provide some information" rather than answering the informational question, a plausible side-effect of the refusal/redirect examples' phrasing over-generalizing to a legitimate in-domain question. **B3** shows a minor brand-voice regression ("our Customer Support team" in v1 -> "the bank" in v2). B1, B7, B8 stayed stable.

Task 5 -- Three-Way Evaluation and Verdict (2 marks)

(Source: `notebooks/05_evaluation_verdict.ipynb`, `results/task5_*`)

Three-way (A/B/C) comparison on the frozen 8-prompt benchmark: full verbatim responses in `results/task5_three_way_comparison.csv`.

Disclosed manual rubric (0-2 per criterion: 0=fails, 1=partially meets, 2=fully meets; `hallucination` inverted, 2=no fabricated facts):

Criterion	Meaning
`domain_correctness`	Is the banking information factually right for NovaBank's stated process?
`instruction_format_adherence`	Did the response follow any explicit formatting instruction (esp. B4)?
`completeness`	Does it fully answer, including every sub-part (esp. B5's 2-step reasoning)?
`hallucination`	Inverted: 2 = no fabricated facts, 0 = fabricated facts present

Note on the rubric's `hallucination` criterion for a fictional bank: since NovaBank has no real ground truth, plausible fictional specifics (an invented phone number, a URL) consistent with the persona were *not* penalised -- only internal contradictions or unsupported specific claims presented with false confidence (e.g. incorrect arithmetic, an unsourced "60-day" policy claim) were scored down.

Per-system totals (`results/task5_rubric_totals.csv` , max 16/criterion, 64/system):

System	domain_correctness	instruction_format_adherence	completeness	hallucination	**Grand total**
A (base SLM)	13	16	13	12	**54 / 64**
B (reference LLM)	16	16	16	16	**64 / 64**
C (fine-tuned v2)	12	16	12	15	**55 / 64**

Automatic metric (A vs. C): ROUGE-L F1 over the full 26-example held-out test set: A = **0.1774**, C = **0.3108**, delta = **+0.1335** (`results/task5_automatic_metric.json`). A real, substantial surface-overlap gain -- C's phrasing/structure genuinely converged toward NovaBank reference responses. **Limitation:** ROUGE-L measures surface lexical/sequence overlap, not semantic correctness -- a response can be factually right but differently phrased (scoring low) or copy reference phrasing while being subtly wrong (scoring high); it should be read alongside the rubric scores above, not in isolation.

Honest error analysis (>=2 cases where C loses to B):

1. **Prompt B6** (out-of-scope): C writes a full cookie recipe; B correctly declines. **Diagnosis -- data problem, not capacity.** C is demonstrably capable of a short refusal (base SLM A produced exactly that at baseline, pre-fine-tuning). v2's training removed that behavioral prior: only 15/321 (4.7%) examples model refusal against 306 that all model "answer fully," so gradient updates favored the dominant pattern. Confirmed by Task 4 Step 8: the same targeted fix measurably failed to change this behavior between v1 and v2.

2. **Prompt B2** (home loan documents): C asks the user for personal details instead of listing required documents; B gives a direct 7-item checklist. **Diagnosis -- training/data problem, not capacity.** v1 answered this exact prompt correctly (Notebook 03's demo, Notebook 04 Step 1). The regression is best explained as the newly added refusal/redirect examples' style over-generalizing past their intended scope, to a legitimate in-domain question that should have been answered directly. With only 321 total examples, the training set doesn't sharply separate "redirect because out-of-scope" from "ask a clarifying question," so C partially conflated the two intents.

Verdict. B scores a perfect 64/64; A scores 54/64; C scores 55/64 -- edging out A overall, but unevenly: C closes real ground on reasoning (B5: A 4/8 -> C 7/8, matching B's correct arithmetic) and brand-voice consistency, but loses ground A already had on two prompts (B2: 7->4, B6: 8->4), both concrete diagnosed regressions rather than noise. Given Task 3's efficiency numbers (0.24% trainable parameters, <6GB peak training memory -- a fraction of what full fine-tuning or running a 7B reference model would cost), C is a reasonable choice for **high-volume, template-like, in-domain intents with clear training coverage** (password resets, card blocking/cancellation, balance-and-transfer reasoning), where NovaBank could route routine support volume to a private, cheap, on-premise model instead of an external API. It is **not** yet a safe drop-in replacement for B: the two confirmed regressions are exactly the kind of unpredictable behavior a bank cannot ship unreviewed -- an out-of-scope request should never produce a full unrelated answer, and a factual question should never be met with a non-answer. Both are diagnosed as data problems, not capacity problems -- the encouraging finding, since the fix is better-balanced training data, not a bigger base model. Until that iteration happens, a hybrid routing setup (C for known-good intents, B or a human agent for the rest) is the practical deployment path this evidence supports.

Appendix A -- Virtual Lab Screenshots

Not applicable in the usual sense -- see Appendix C. The BITS Virtual Lab instance assigned for this course had no NVIDIA GPU (`lspci` showed only the standard AWS EC2 virtual display adapter, no CUDA device) and its root disk was at 100% capacity (172K free of 32GB) with no usable space to install the ML dependency stack. Screenshots of these two blocking conditions on the Virtual Lab, taken before switching to local execution, are attached separately per the instructions in Appendix C.

Appendix B -- References

1. Bitext Innovations International, S.L. *Retail Banking LLM Chatbot Training Dataset* (2024). Hugging Face Datasets.
2. Qwen Team, Alibaba Cloud. *Qwen2.5 Technical Report* (2024).
3. Hu, E. J., et al. *LoRA: Low-Rank Adaptation of Large Language Models.* arXiv:2106.09685 (2021).
4. Dettmers, T., et al. *QLoRA: Efficient Finetuning of Quantized LLMs.* arXiv:2305.14314 (2023).

Appendix C -- Note on Execution Environment (Deviation from PS2 Section 3)

PS2 Section 3 requires strict execution in the Virtual Lab. This could not be met for Tasks 2-5: the assigned Virtual Lab instance had **no GPU** (verified via `lspci` / `nvidia-smi` -- only a non-GPU AWS EC2 virtual display adapter was present) and its **root filesystem was full** (32GB volume, 172K free, `/home/cloud/.cache/pipenv` and `.cache/pip` alone consuming ~21GB from repeated dependency install attempts, with no room left to install `torch` / `transformers` / `bitsandbytes` or download model weights). Running a 1.5B/7B-parameter model fine-tuning and inference workload on that CPU-only, disk-exhausted instance was not feasible in any practical timeframe (full fine-tuning-scale runs would take on the order of days on CPU, per PS2's own domain requirements). Screenshots of both blocking conditions, taken on the Virtual Lab before the switch, are attached as proof.

Tasks 2-5 were instead executed locally on an **NVIDIA GeForce RTX 4050 Laptop GPU (6GB VRAM)**, using the same pinned dependency versions (`Pipfile` / `Pipfile.lock`), the same `SEED=42` reproducibility discipline, and the same models (`Qwen/Qwen2.5-1.5B-Instruct` , `Qwen/Qwen2.5-7B-Instruct`) specified throughout this report. All notebook outputs, loss curves, saved adapters, and evaluation numbers in this report are real, executed outputs from that local run -- none were fabricated or estimated.